



Global Olympic Games Data Analysis

An End-to-End Portfolio Project Using PostgreSQL & Power BI

Project Overview & Objectives

Core Mission: Transforming millions of rows of historical sports records into interactive, business-ready insights.

Dataset Scope: Cleaned and structured 25 distinct CSV files tracking athletes, regions, events, and medals.

Technical Core: Executed 18 complex SQL query scripts in PostgreSQL to extract trends and event anomalies.

Visualization: Built a high-performance, 4-tab Power BI dashboard for global performance storytelling.

SQL Engineering & Analytics

Advanced Time Intelligence: Developed decade-based mathematical blocks to isolate sports hosting frequencies.

Relational Modeling: Implemented multiple table JOINS linking structural data across games, competitors, and events.

Anomaly Hunting: Wrote queries targeting specific performance spikes, participation streaks, and unexpected trends.

Optimization: Formatted clean, documented, and reusable PSQL code across 18 distinct analysis modules.

Dashboard Tabs 1 & 2: Overview & Demographics

Olympic Games Overview: High-level KPI cards displaying running totals for game editions, sports variants, and nations.

Growth Trends: Dynamic line charts highlighting expanding participation patterns across historical eras.

Athlete Demographics: Deep dive into physical metrics (age, height, weight distribution) across competitor categories.

Diversity & Representation: Tracked gender distribution ratios changing over more than a century of games.

Dashboard 1



Dashboard 2

Athlete & Sport Demographics

Total Events
757



Average of height
138 kg



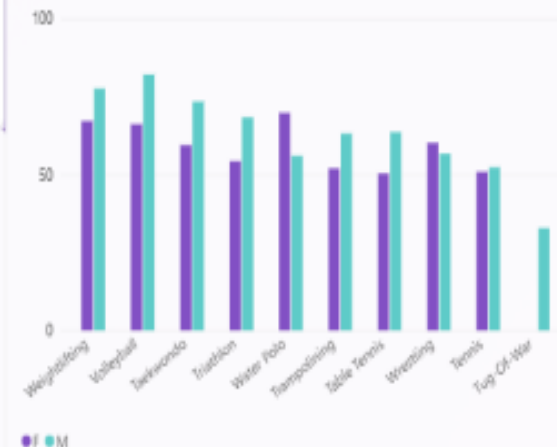
Average of weight
56 kg



Average of age
25.78



Average of weight
by sport_name and gender



gender

☐ F

☐ M

season

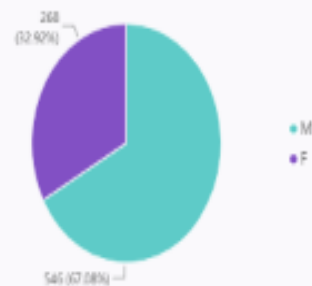
☐ Summer

☐ Winter

region_name

All

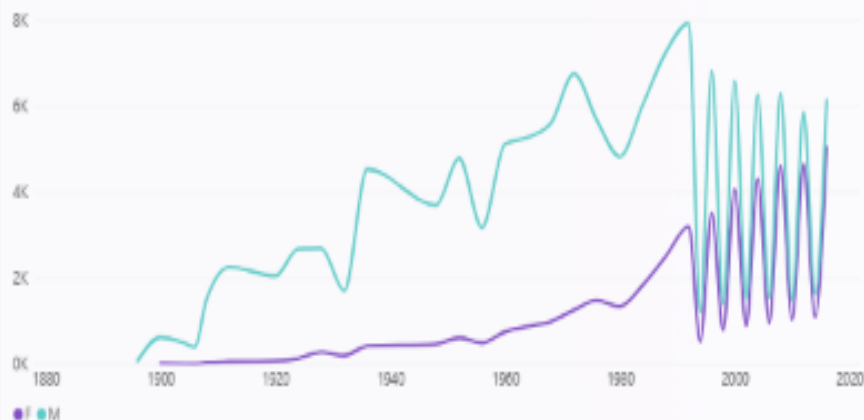
Distribution of Events by gender



Average Athlete Age Profile Over Time



Historical Athlete Participation by gender



Top 10 Sports by total event count



Dashboard Tabs 3 & 4: Medals & Milestones

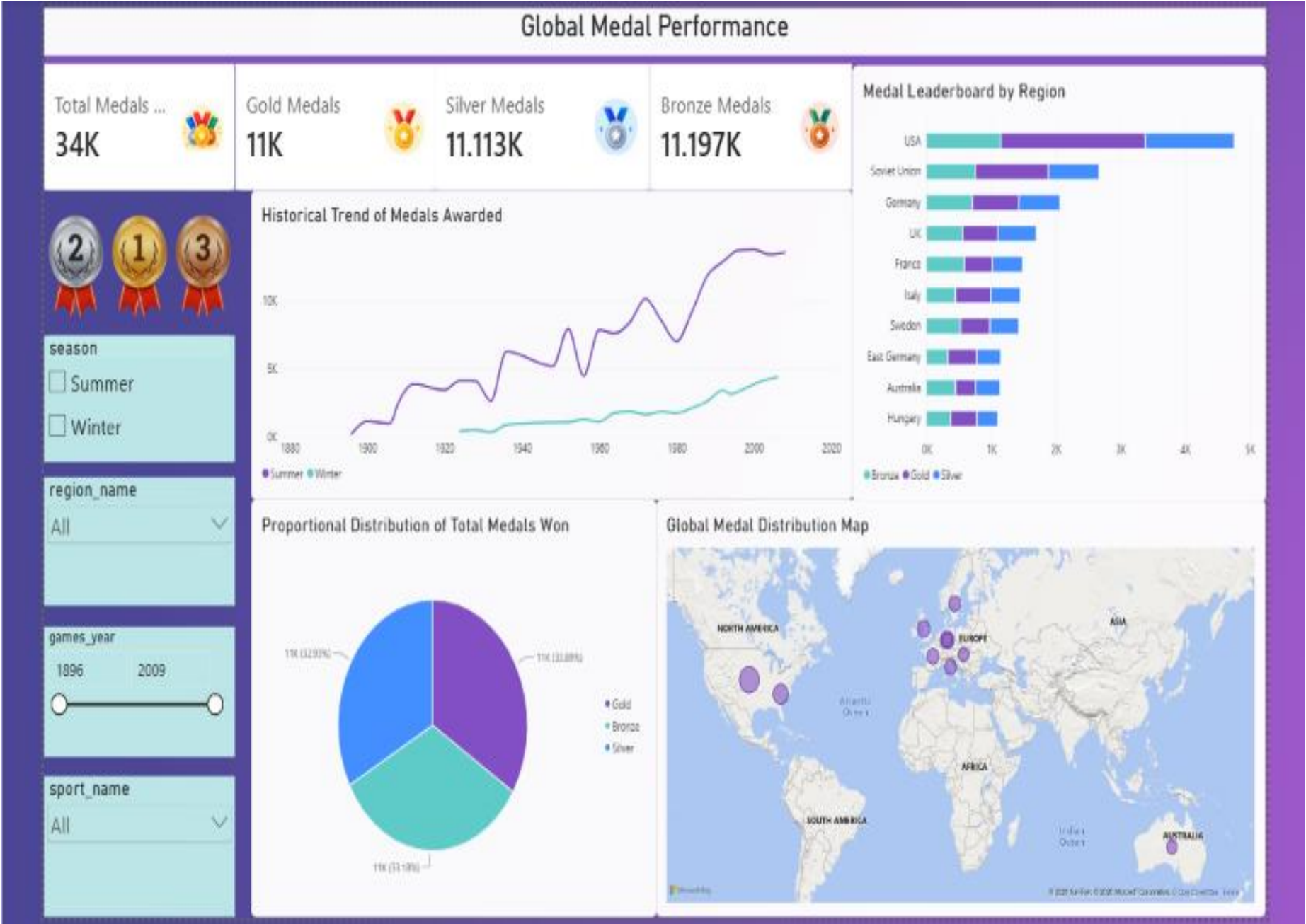
Global Medal Performance: Stacked bar chart leaderboards ranking competitive nations by Gold, Silver, and Bronze splits.

Geographic Mapping: Interactive bubble map showcasing global medal concentrations and regional success hubs.

Anomalies & Event Milestones: Tracked structural changes in competition categories across historical milestones.

Discontinued Sports Analysis: Treemaps visualizing the lifespans of historic sports categories (e.g., Tug-of-War).

Dashboard 3



Dashboard 4

Anomalies & Event Milestones

Total Sports Varieties
231



Participating Nations
230



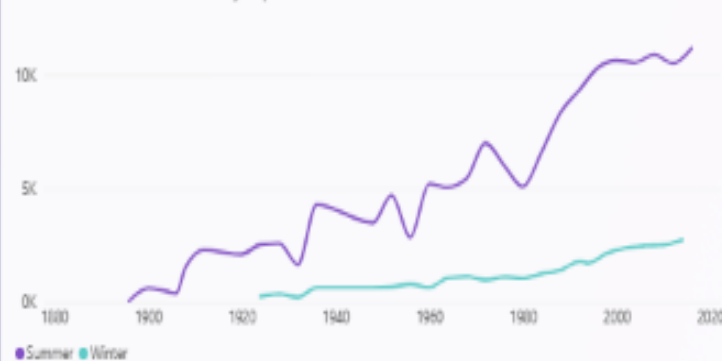
Discontinued Sports
32



Distribution of Sports Varieties by Historical Debut Year



Historical Growth of Olympic Events Over Time



season

☐ Summer

☐ Winter

games_year

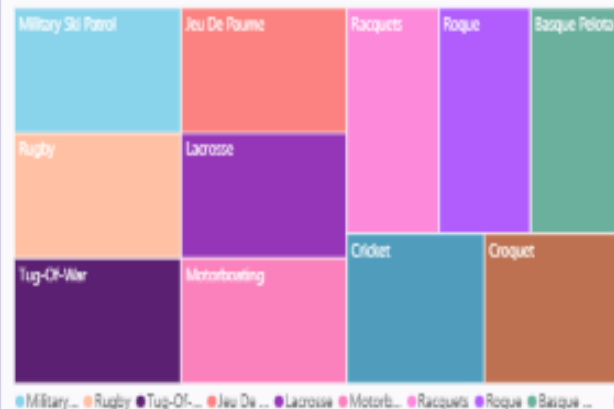
1896

2016

sport_name

All

Historical Timeline of Discontinued Sports



Total Event Count per Game Year



Key Takeaways & Technical Summary

Data Validation: Successfully validated dense SQL metrics against the final interactive reporting layer.

Historical Insights: Visualization confirms massive expansion in global sports diversity and parity after the mid-20th century.

Full Stack Delivery: Seamlessly bridged flat-file CSV ingestion with robust database modeling and frontend BI layout.

Thank you!